

Module 15: Power BI Service & Cloud Reporting

Overview

Creating reports in Power BI Desktop is only part of the Business Intelligence lifecycle. Organizations need a way to publish, share, secure, collaborate on, and refresh reports for business users.

This is where Power BI Service comes in.

Power BI Service is Microsoft's cloud-based Business Intelligence platform that enables report publishing, dashboard creation, collaboration, scheduled refreshes, governance, and enterprise-scale analytics.

This module covers the complete Power BI cloud ecosystem, including Workspaces, Dashboards, Apps, Data Refresh, Sharing, Security, and Deployment Pipelines.

Learning Objectives

After completing this module, you will be able to:

- Understand Power BI Service architecture
 - Publish reports to the cloud
 - Create and manage workspaces
 - Build dashboards in Power BI Service
 - Configure scheduled refreshes
 - Share reports securely
 - Understand Power BI Apps
 - Manage permissions and access
 - Understand deployment pipelines
 - Learn Power BI governance concepts
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1 Introduction to Power BI Service

What is Power BI Service?

Power BI Service is the cloud-based platform used to:

- Publish reports
- Share dashboards
- Collaborate with teams
- Schedule refreshes
- Manage security
- Distribute analytics

Access Method

Power BI Service is accessed through a web browser.

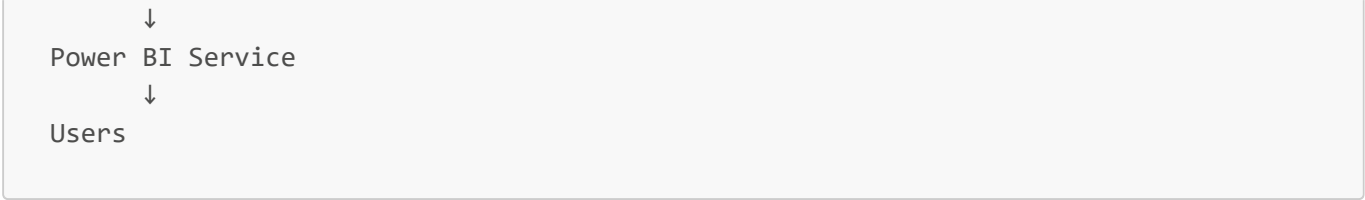
Primary Purpose

Transform personal reports into enterprise reporting solutions.

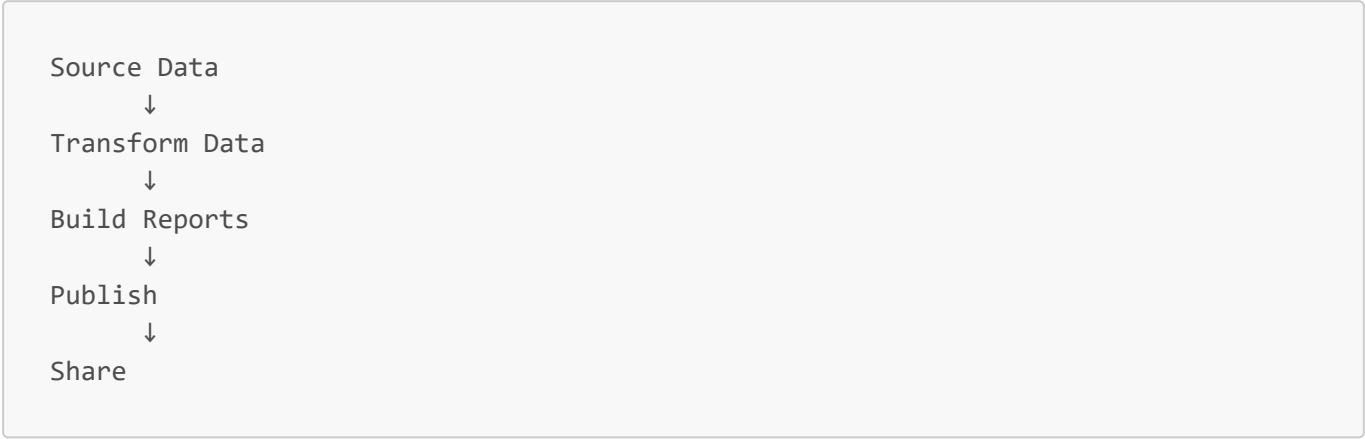
2 Power BI Architecture

High-Level Architecture

```
graph TD
    A[Data Sources] --> B[Power BI Desktop]
```



Data Flow



Components

Power BI Desktop

Development environment.

Power BI Service

Cloud platform.

Power BI Mobile

Report consumption.

Gateway

On-premise connectivity.

3 Power BI Desktop vs Service

Feature	Desktop	Service
Build Reports	Yes	Limited
Data Modeling	Yes	No

Feature	Desktop	Service
DAX Development	Yes	No
Publish Reports	Yes	No
Share Reports	No	Yes
Schedule Refresh	No	Yes
Collaboration	No	Yes
Dashboard Creation	Limited	Yes

Simple Rule

Desktop = Development

Service = Distribution



Power BI Workspaces

What is a Workspace?

A Workspace is a collaborative environment where reports, dashboards, datasets, and apps are stored.

Purpose

Workspaces help teams:

- Collaborate
 - Organize content
 - Manage permissions
-

Example

Sales Team Workspace

Contains:

- Sales Reports
 - Sales Dashboards
 - Sales Datasets
-



Workspace Roles

Power BI provides multiple workspace roles.

Admin

Full control.

Can:

- Add users
- Remove users
- Manage permissions

Member

Can edit content.

Contributor

Can publish and update content.

Viewer

Can only view content.

Role Hierarchy

```
Admin
↓
Member
↓
Contributor
↓
Viewer
```

Publishing Reports

What is Publishing?

Publishing uploads a Power BI report from Desktop to Power BI Service.

Steps

Step 1

Open Power BI Desktop.

Step 2

Save Report.

Step 3

Select:

Home → Publish

Step 4

Choose Workspace.

Step 5

Upload Report.

Result

Report becomes available in Power BI Service.

Datasets & Semantic Models

What is a Dataset?

A Dataset contains:

- Imported Data
 - Relationships
 - Measures
 - Calculations
-

Modern Terminology

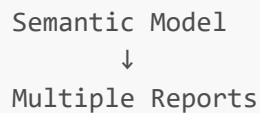
Microsoft increasingly refers to datasets as:

Semantic Models

Why Important?

Reports connect to semantic models.

Architecture



```
graph TD; A[Semantic Model] --> B[Multiple Reports]
```

Benefits

Single source of truth.

8 Dashboards in Power BI Service

What is a Dashboard?

A dashboard is a collection of visual tiles from one or more reports.

Characteristics

- Single Page
 - Interactive
 - High-Level Overview
-

Example

Executive Dashboard

Contains:

- Revenue KPI
 - Profit KPI
 - Sales Trend
 - Top Products
-

9 Tiles & Dashboard Design

What is a Tile?

A tile is an individual visual pinned to a dashboard.

Examples

- KPI Card
 - Line Chart
 - Map
 - Table
-

Pinning a Visual



```
graph TD; Report --> PinVisual[Pin Visual]; PinVisual --> Dashboard;
```

Report
↓
Pin Visual
↓
Dashboard

Benefits

Allows consolidation of insights from multiple reports.

Power BI Apps

What is a Power BI App?

An App is a packaged collection of:

- Reports
- Dashboards
- Semantic Models

Distributed to users.

Benefits

Centralized Distribution

One location for content.

Controlled Access

Users consume without editing.

Professional Deployment

Enterprise-friendly.

1 1 Data Refresh

Why Refresh?

Source data changes continuously.

Reports must remain current.

Types of Refresh

Manual Refresh

User initiates refresh.

Scheduled Refresh

Automatic updates.

Examples:

```
Daily
Hourly
Weekly
```

Incremental Refresh

Refreshes only changed data.

Benefits

- Faster updates
 - Better performance
-

1 2 Gateways

What is a Gateway?

A gateway securely connects on-premise systems to Power BI Service.

Why Needed?

Power BI Service cannot directly access internal databases.

Types

Personal Gateway

Single-user use.

Standard Gateway

Enterprise use.

Multiple users supported.

Architecture

```
graph TD; A[SQL Server] --> B[Gateway]; B --> C[Power BI Service];
```

SQL Server
↓
Gateway
↓
Power BI Service

Sharing Reports

Sharing Options

Direct Sharing

Share report with users.

Workspace Access

Users access workspace content.

App Distribution

Publish apps to groups.

Embedded Analytics

Integrate reports into applications.

Permissions & Security

Why Security Matters

Business reports often contain sensitive information.

Permission Types

View

Read-only access.

Build

Create reports using datasets.

Edit

Modify content.

Reshare

Allow sharing with others.

Principle

Least Privilege Access

Grant only necessary permissions.

Row-Level Security (Overview)

What is RLS?

Row-Level Security restricts data visibility based on users.

Example

Manager

Sees all regions.

Salesperson

Sees only assigned region.

Benefits

Protects sensitive data.

Detailed Implementation

Covered in Module 16.

Collaboration Features

Power BI Service supports collaboration.

Comments

Users discuss report insights.

Subscriptions

Receive reports via email.

Alerts

Notifications when KPIs exceed thresholds.

Teams Integration

Integrates with Microsoft Teams.

Deployment Pipelines

What is a Deployment Pipeline?

A deployment pipeline manages report promotion across environments.

Stages

```
graph TD;
  Development --> Test;
  Test --> Production;
```

Benefits

Quality Assurance

Test before release.

Controlled Deployment

Reduces production risks.

Enterprise Governance

Supports large teams.

1 **8** Governance & Administration

Governance

Governance ensures Power BI usage follows organizational standards.

Areas

Security

Access control.

Compliance

Regulatory requirements.

Data Quality

Reliable reporting.

Monitoring

Usage tracking.

Administrative Responsibilities

- User Management
 - Capacity Management
 - Audit Logs
 - Workspace Governance
-

Licensing Overview

Power BI licensing affects sharing capabilities.

Power BI Free

Individual use.

Power BI Pro

Collaboration and sharing.

Power BI Premium Per User (PPU)

Advanced enterprise features.

Power BI Premium Capacity

Large-scale enterprise deployment.

General Rule

Most organizations use:

Pro

or

Premium

2 0 Best Practices

Use Workspaces Properly

Separate teams and projects.

Schedule Refresh Carefully

Avoid unnecessary refresh frequency.

Use Apps for Distribution

Provides better governance.

Implement Security Early

Protect sensitive data.

Monitor Usage

Identify adoption trends.

2 1 Common Mistakes

Sharing Without Security Planning

Creates risks.

Excessive Workspace Creation

Causes management issues.

Ignoring Refresh Failures

Leads to stale reports.

No Governance Strategy

Creates reporting chaos.

Publishing Directly to Production

Increases risk.



Key Terminologies

Term	Definition
Power BI Service	Cloud-based BI platform
Workspace	Collaboration environment
Dashboard	Collection of visual tiles
Tile	Individual dashboard visual
Semantic Model	Dataset containing business logic
App	Packaged reporting solution
Gateway	Secure on-premise connector
Refresh	Updating data
Deployment Pipeline	Environment promotion process
RLS	Row-Level Security



Practice Exercises

Exercise 1

Create a Workspace.

Exercise 2

Publish a report from Power BI Desktop.

Exercise 3

Create a Dashboard using pinned visuals.

Exercise 4

Schedule a data refresh.

Exercise 5

Research Power BI licensing options.



Mini Project

Cloud Reporting Solution

Objective

Publish and distribute a Sales Dashboard.

Tasks

Workspace

Create:

Sales Analytics Workspace

Publish

Upload report.

Dashboard

Pin:

- Revenue KPI
 - Profit KPI
 - Sales Trend
-

App

Create:

Sales Analytics App

Security

Assign:

- Viewer
- Contributor

roles.

Deliverables

1. Published Report
 2. Dashboard
 3. Workspace Setup
 4. App Distribution Plan
-



Module Summary

In this module, you learned:

- ✓ Power BI Service Architecture
- ✓ Workspaces
- ✓ Workspace Roles
- ✓ Publishing Reports
- ✓ Semantic Models
- ✓ Dashboards
- ✓ Tiles
- ✓ Power BI Apps
- ✓ Data Refresh
- ✓ Gateways
- ✓ Sharing & Collaboration
- ✓ Security & Permissions
- ✓ Deployment Pipelines
- ✓ Governance & Administration

Power BI Service transforms standalone reports into enterprise reporting solutions. Understanding the Service layer is essential for sharing insights, managing security, and supporting business users at scale.



Next Module

Module 16: Row-Level Security (RLS) & Data Security

Topics Covered:

- Data Security Fundamentals
- Row-Level Security (RLS)
- Static RLS
- Dynamic RLS
- USERPRINCIPALNAME()
- Security Roles
- Testing Security
- Object-Level Security (OLS)
- Data Governance
- Enterprise Security Best Practices